Communications Module Model 3: Secondary Network Link Interface Reassignment

for SM-2000 or EXM-2000

Excerpt from an installation procedure. Selected sections, figures, and one table are shown; step, figure, and table numbers are kept from the full procedure. Company names, product brand, document numbers, part codes, and internal links have been removed.

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1. General Information

1.1 Purpose

This section describes the procedure to reassign and move secondary Network Link Interface (NLI) pairs (link sets) on a local Switching Module-2000 (SM-2000) and an Extended Switching Module-2000 (EXM-2000) with a Transmission Rate Converter Unit, version 2 or 3 (TRCU2 or TRCU3) in an office equipped with a Communications Module Model 3 (CM3) from one Time Multiplexed Switch Fabric Pair (TMSFP) cabinet to a growth TMSFP cabinet.

SCOPE

This section supports the reassignment and movement of secondary NLI link set one block at a time for an SM-2000 or an EXM-2000 from one TMSFP cabinet to a growth TMSFP cabinet. A "block" is defined as a minimum of one link set of a single SM-2000/EXM-2000.

1.4 Safe Stop Points

The steps in the procedure labeled "Safe Stop Point" are points where progress in the execution of the procedure may be suspended temporarily without causing degradation in the operation of the switch. Execution of the procedure can be halted at one of these points only if all prior steps have been successfully completed.

1.5 Hardware Description

Secondary NLI paddle boards are used to support additional Network Time Slots terminated by Network Control and Timing/2 (NCT/NCT2) links between the CM3 and the TSIU4/TSIU4-2. See Figure 1 for the TSIU4 and Figure 2 for the TSIU4-2.

CM3 Side 0 and Side 1 physical arrangement is configured back-to-back (Figure 3 and Figure 4). The TMSU4 (Figure 5) and the CMU2 (Figure 6 and Figure 10) contain Optical Paddle Boards (OPB) where NCT/NCT2 links from the NLIs terminate (Figure 7). Conventional engineering of additional CM3 cabinets that terminate secondary NLIs must be placed to the right of the original CM3 cabinet (Figure 8). The expansion CM3 cabinets contain three TMSU4 shelves and one CMU2 shelf. The expansion CMU2 shelf (Figure 9 and Figure 11) contains a subset of circuit boards of the base CMU2.

1.7 Prerequisite Conditions

1. If the diagnostic results from a recent (1 or 2 previous day period) Routine Exercise (REX) are clean (e.g. fault free) for both sides of each MCTSI and the ONTC in which this reassignment is planned, you can continue with this procedure. If REX produced a problem result(s) or REX has not been run recently, a diagnostic result of All Tests Pass (ATP) for both sides of each MCTSI and ONTC must be achieved before continuing with this procedure.
2. The new TMSFP cabinet(s) must be identified.
3. The original TMSFP cabinet(s) must be identified.
4. This office must be on the required software release or later, with:

- the required software update with CFT at00 (for offices equipped with ASM).
- the required software update with CFT lt00 (for offices without ASM).

5. The Advance Preparation Section of this procedure must be completed prior to beginning the actual NLI Link Reassignment.

CAUTION: Do NOT migrate all secondary NLIs of the same SM at one time. Migration must be done on a per link set basis across all SMs that are affected. A "block" in this procedure refers to performing migration of one link set per SM across all affected SMs with the limitation of how many blocks can be migrated in each maintenance window required for the entire job.

2. Advance Preparation

NOTE: The steps in the Advance Preparation section of this procedure should be performed 1 week prior to the scheduled NLI Reassignment. These advance preparation steps entail identifying the Secondary NLIs that will be reassigned and identifying the order or sequence in which they will be reassigned. The Installer must complete Table B, which identifies the Secondary NLIs being reassigned and then complete Table C, which identifies the order of the NLI reassignment.

NOTE: The term "block" is used throughout this procedure. A block is a group of NLI Sets that must be moved together. Each copy of Table C is a block. NLI Sets in a block :

- Must be from the same original TMSFP migrating to the same New TMSFP.
- Must only include 1 NLI Set from each SM (NLI Set Numbers may be different).

2.1 Identify Secondary NLIs to be migrated

2.1.1 Identify all SM2000s/EXM2000s in the office.

2.1.2 At the MCC, type and enter:

```
OP:SYSSTAT,UCL;

Response: OP SYSSTAT COMPLETED
```

NOTE: The output of the OP:SYSSTAT command will identify the SM configuration. SM2000's are denoted by the letter `K' in the first column.

2.1.3 To populate the Original column (left side) of Table B, access the NLI Summary Page. At the MCC, type and enter:

CMD: 1200,X

```
Where:  X = SM number determined in Step  2.1.2 .

Response:  NLI Summary Page Displayed.
```

2.1.4 Use the 1200 NLI Summary page to determine the 1201 NLI/TMSLNK SET X MCC page and set number for the identified NLIs being reassigned.

2.1.5 Use the 1201 NLI/TMSLNK SET X MCC page to match reassignment NLIs to the original TMSFP and the original TMSLNK set. At the MCC, type and enter:

CMD: 1201,X

```
Where:  X = NLI set number

Response:  NLI/TMSLNK Set X Page Displayed.
```

2.1.6 To populate the New column (right side) of Table B, use the supplied GRCV 18.16 information from Section 1.6 to identify those Link Sets being reassigned that are associated with the SM numbers listed in Step 2.1.2 .

1. Were any discrepancies found between the Link Sets identified in Table B with the actual office equippage?

- If YES, Contact Regional Engineering for support. Do NOT continue until the discrepancy has been resolved.
- If NO, continue with Step 2.1.7 .

2.1.7 Populate Table C: Determine SM NLI Link Set Migration order.

1. Make 10 copies of Table C.

2. Obtain a completed set of Table B (Link Sets 1 through 11). Link Set 1 is the primary NLI set and is not migrated.
3. Using the completed Table B, populate Table C in the following manner:
    a. In Table B, beginning with TMSFP 0 column for Link Set 2, locate the first group of TMS Links identified for reassignment.
    b. Record the data for Link Set 2 in the first empty row of Table C.
    c. Record data for Link Sets (3-11) on TMSFP 0 in the successive rows after Link Set 2.
    d. Next, beginning with the TMSFP 1 column in Table B for Link Set 2, locate the first group of TMS Links identified for migration.
    e. Record the data for Link Set 2 in the first empty row of Table C.
    f. Record data for Link Sets (3-11) on TMSFP 1 in the successive rows after Link Set 2.
    g. Next, beginning with the TMSFP 2 column in Table B for Link Set 2, locate the first group of TMS Links identified for migration.
    h. Record the data for Link Set 2 in the first empty row of Table C.
    i. Record data for Link Sets (3-11) on TMSFP 2 in the successive rows after Link Set 2.
    j. Finally, beginning with the TMSFP 3 column in Table B for Link Set 2, locate the first group of TMS Links identified for migration.
    k. Record the data for Link Set 2 in the first empty row of Table C.
    l. Record data for Link Sets (3-11) on TMSFP 3 in the successive rows after Link Set 2.

3. Preliminary

The steps in this Preliminary section, should be done at least 2-4 hours prior to the start of the maintenance window.

WARNING: TMS Links should not be left in SPECIAL GROW for extended periods of time. The TMS Link state can be visually determined by inspecting MCC pages 1221-1228 for Side 0 of the specified TMSFP number. The Side 1 TMS Link state can be determined by inspecting MCC pages 1231-1238 for the specified TMSFP number. A TMS Link is marked SPECIAL GROW via RC/V 17.12 when the NLI is in the GROW state. Then when the NLI is marked OPERATIONAL via RC/V 18.16, the TMS Link can be marked OPERATIONAL via RC/V 17.12. Under no circumstances should a TMS Link be left in SPECIAL GROW beyond the current installation shift.

If the NLI fails to test or link, do the following:

1. Put the TMS Link back in SPECIAL GROW via RC/V 17.12.
2. Put the NLI back in GROW via RC/V 18.16.
3. Put the TMS Link back in GROW via RC/V 17.12.
4. Correct the problem.

3.1 Determine Switch Condition

3.1.1 At the MCC observe the Summary Status Area for a SYS NORM indication.

- If the SYS NORM indicator is highlighted in green, go to Step 3.2.1 .
- If the SYS NORM indicator is not highlighted in green, continue with Step 3.1.2 .

3.1.2 If a SYS NORM indication is not observed, determine the reason why. At the MCC, type and enter:

```
OP:SYSSTAT;

Response:  OP SYSSTAT report printed on the ROP.
```

3.1.3 At the MCC, type and enter:

```
OP:OFFNORM,SM=a;

Where: a = SM-2000 number with secondary NLIs being reassigned

Response: OP OFFNORM report printed on the ROP.
```

3.1.4 At the MCC, type and enter:

```
1010,a

Where: a = SM-2000 number with secondary NLIs being reassigned

Response:  The SM Status MCC page is displayed.
```

3.1.5 Examine the SM Status MCC page for the identified SM-2000, and verify that there are no alarm indications for the SM-2000, and that the MCTSI and the SM-2000 peripherals are all in service.

3.1.6 Evaluate the reports and the SM Status page to determine if system status is acceptable to continue. Specific attention is required for areas that can disrupt call processing. If any problems exist that could impact the execution of the procedure or call processing they must be resolved prior to continuing.

4. NLI Work

4.2 Set ONTCCOM Side 0 to be Active Major

CAUTION: If the office network and timing complex (ONTC) has a degraded condition, clear the problem(s) first. Do NOT continue until the degraded condition has been cleared. Do NOT switch the ONTC unconditionally.

4.2.1 At the MCC, type and enter:

```
1209
```

```
Response: Page for ONTC 0 & 1 displayed
```

4.2.2 Is ONTCCOM side 0 ACT MAJOR?

- Yes - skip to Step 4.3 .
- No - Continue with Step 4.2.3 .

4.2.3 At the MCC, type and enter:

```
SW:ONTC;
```

```
Response: SW ONTC COMPLETED
```

4.3 Remove Side 1 NLI Link Set(s) From Service

4.3.1 At the MCC, type and enter:

```
1200,a
```

```
Where: a = SM-2000 number with secondary NLI being reassigned
```

```
Response:
    SM NLI page displayed
```

4.3.2 At the MCC, type and enter:

```
RMV:NLI=a-bb-1;
```

```
Where: a = SM-2000 number with secondary NLI being reassigned
```

```
    bb = NLI Even number
```

```
Response:
    RMV NLI=a-bb-1 COMPLETED
```

4.3.3 At the MCC, type and enter:

```
RMV:NLI=a-bb-1;
```

```
Where: a = SM-2000 number with secondary NLI being reassigned
```

```
    bb = NLI Odd number
```

```
Response:
    RMV NLI=a-bb-1 COMPLETED
```

4.3.4 Repeat Step 4.3.1 through Step 4.3.3 for each SM in the block.

4.4 Set ONTCCOM Side 1 to be Major

4.4.1 At the MCC, type and enter:

```
14x

Where: x = 1, 2, 3, or 4

Response:
    SM STATUS SUMMARY page displayed

Response: Page for ONTC 0 & 1 displayed
```

4.4.2 Inspect each SM for CCSLK off normal alarms.

4.4.3 Repeat Step 4.4.1 and Step 4.4.2 for all 14x pages.

4.4.4 If CCS functionality is impaired on any SM, access MCC page 1532,x (Where x = GSM number) to investigate the affected links. The individual links can be accessed through the appropriate 1533 page.

CAUTION: All impacted F-Links must be restored prior to the removal of any additional NLIs.

4.4.5 At the MCC, type and enter:

```
1209

Response: Page for ONTC 0 & 1 displayed
```

4.4.6 At the MCC, type and enter:

```
SW:ONTC,UCL;

Response: SW ONTC COMPLETED
```

NOTE: This is the only unconditional ONTC switch in the procedure. It is needed because the Side 1 NLIs were removed in Step 4.3. Do not use it unless Steps 4.4.1 through 4.4.4 show no CCS impairment and the ONTC has no other degraded condition.

4.5 Remove Side 0 NLI Link Set(s) From Service

4.5.1 At the MCC, type and enter:

```
1200,a

Where: a = SM-2000 number with secondary NLI being reassigned

Response:
    SM NLI page displayed
```

4.5.2 At the MCC, type and enter:

```
RMV:NLI=a-bb-0,REMAP;

Where: a = SM-2000 number with secondary NLI being reassigned

    bb = Even NLI number

Response:
    RMV NLI=a-bb-0 COMPLETED
```

NOTE: The time required for this command to complete could take approximately 16 minutes.

CAUTION: The NLI removals should be spaced apart to allow ROP monitoring for messages indicating that CCS functionality has been impaired. The message below will be output: ** REPT GLOBAL FUNCTIONALITY IMPAIRED TYPE=SIG GTWY GSM=# CCS_LNK If the following message is seen, access MCC page 1532,x (Where x = GSM number) to investigate the affected links. The individual links can be accessed through the appropriate 1533 page. All impacted F-Links must be restored prior to the removal of any additional NLIs.

4.5.3 At the MCC, type and enter:

```
RMV:NLI=a-bb-0,REMAP;

Where: a = SM-2000 number with secondary NLI being reassigned

    bb = Odd NLI number

Response:
    RMV NLI=a-bb-0 COMPLETED
```

NOTE: The time required for this command to complete could take approximately 16 minutes.

4.5.4 Repeat Step 4.5.1 through Step 4.5.3 for each SM in the block. See Note below.

NOTE: The system can queue up to 16 NLI removals, but space the removals as described in the CAUTION after Step 4.5.2. Use the OP:DMQ,CM command to determine the number of RMV NLIs in queue.

4.6 Update Side 0 TMS Links To Special Grow

4.6.1 Access Recent Change View 17.12 in the Update mode.

4.6.2 Enter the following key attributes for the NLI:

RCV 17.12 KEY ATTRIBUTES

*1. SIDE	0
*2. TMSFP NUM	a
*3. LINK	b

```
Where: a = original TMSFP number

    b = TMS link number recorded in Table C for the NLI being reassigned.
```

4.6.3 Change the EQSTAT attribute from O (OPERATIONAL) to S (SPECIAL GROW)

4.6.4 Update the form.

```
Response: updating......FORM UPDATED
```

4.6.5 Repeat Step 4.6.2 through Step 4.6.4 for every set of TMS links being reassigned.

5. Migrate NCT/NCT2 Links

5.3 Connect Network Control and Timing (NCT/NCT2) Links For New TMSFP Side 0

5.3.1 Use Table C to find the SM Link sets Original TMS Link numbers and New TMS Link numbers. Use Table A and the figure associated with the New TMSFP to determine the Original and New location.

5.3.2 On Side 0 of the Original TMSFP's cabinet, remove the identified OPB from the frame, leaving the fibers connected.

5.3.3 Move all the affected OPBs to their New TMSFP location per Table C.

6. Diagnose and Restore Migrated NLI Sets

NOTE: The following activity of diagnosing the NLIs can be done concurrently across all SMs.

6.1 Diagnose Each Growth NLI on Side 0

6.1.1 At the MCC, type and enter:

```
1200,x

Where: x = Growth SM-2000

Response: DLI/NLI SUMMARY page
```

6.1.2 At the MCC, type and enter:

```
DGN:NLI=a-bb-0,RAW,GROW,TLP;

Where: a = Growth SM-2000 number

    bb = NLI number

Response:
    DGN NLI=a-bb-0 COMPLETED ATP
```

6.1.3 Repeat Step 6.1.2 for the other NLI of the side 0 pair.

6.3 Update Growth NLI Set to Operational

6.3.1 Access Recent Change View 18.16 in the Update mode.

6.3.2 Enter the following key attributes for the NLI:

RCV 18.16 KEY ATTRIBUTES

*1. SM	_a
*2. EVEN NLI	__b

```
Where: a = SM-2000 number with secondary NLI being reassigned

    b = NLI number being reassigned
```

6.3.3 Change the EQSTAT attribute from G (GROW) to O (OPERATIONAL).

6.3.4 Update the form.

```
Response: updating......FORM UPDATED
```

6.3.5 Quit the update of Recent Change View 18.16.

6.6 Verify TMS Links

6.6.1 Verify that no TMS links are in the SGRO state. At the MCC, type and enter:

```
OP:CFGSTAT,CM,SGRO;

Response: The CM configuration status is printed.
```

6.6.2 Are any TMS links in the SGRO state?

- If Yes go to Step 6.6.3.
- If No go to Step 6.7.

6.6.3 Determine if the TMS in the SGRO state is part of this growth job (Check GRCV 17.12 forms)

- If the TMS Link is part of this growth job, go to Step 6.4 to change Side 0 TMS Links to "O" or Step 6.5 to change Side 1TMS Links to "O".

- If the TMS Links are not part of this growth job, Remove the parent TMSFP from service, and update the TMS link to "G".

6.6.4 Repeat Step 6.6.3 for each TMS LInk in the SGRO state.

6.6.5 Perform Step 6.6.1 again to verify that no TMS Links are in the SGRO state.

7. Return to Normal Conditions

7.11 Control Decision - (More Blocks to Migrate)

7.11.1 Are there more blocks to migrate, per Table C ?

- YES - During the next available Maintenance window, begin at Paragraph 3 for the next block to be migrated as indicated by Table C.
- NO - Make ODD backup tapes per local procedures. This completes the reassignment of secondary NLIs.

Figures

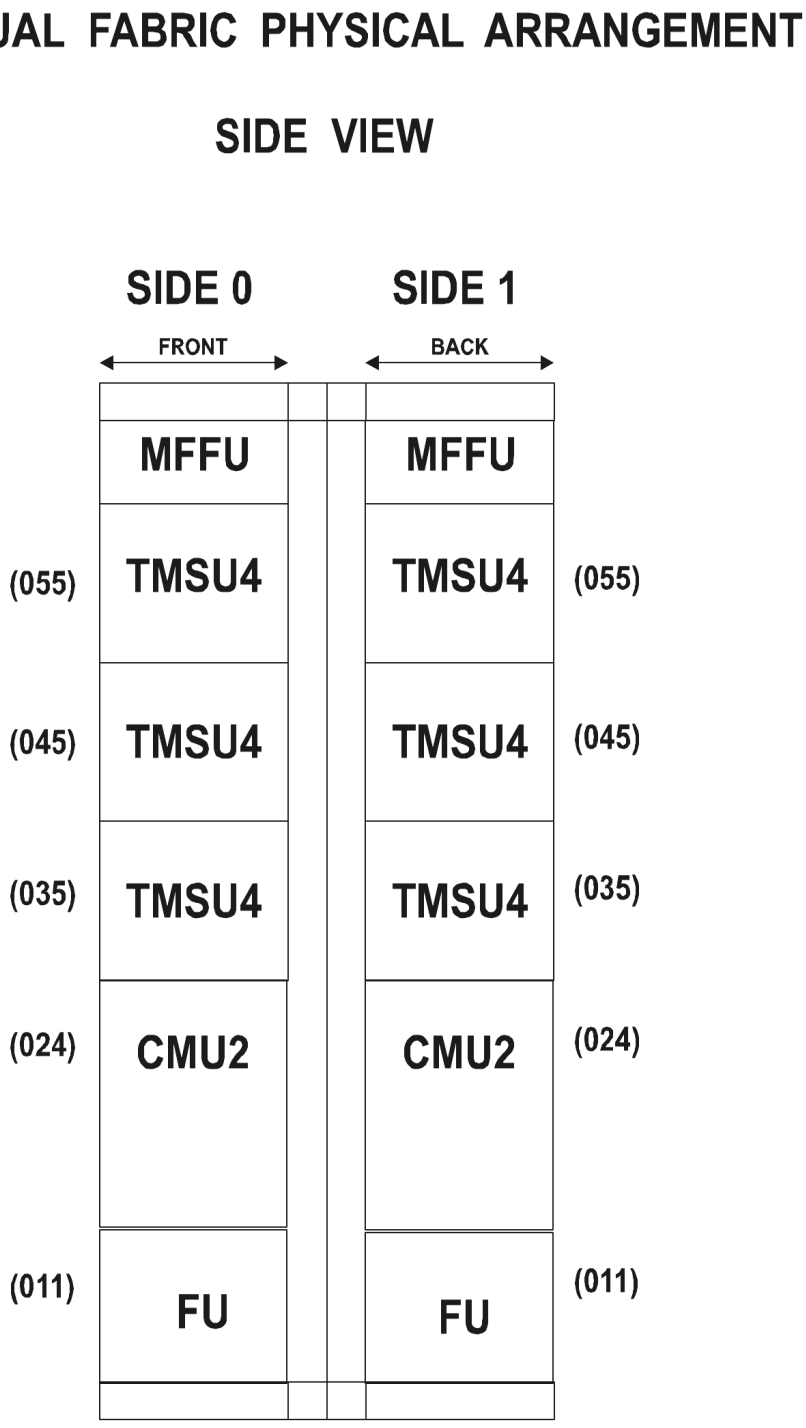


Figure 3. CM3 Dual Fabric Physical Arrangement - Side View

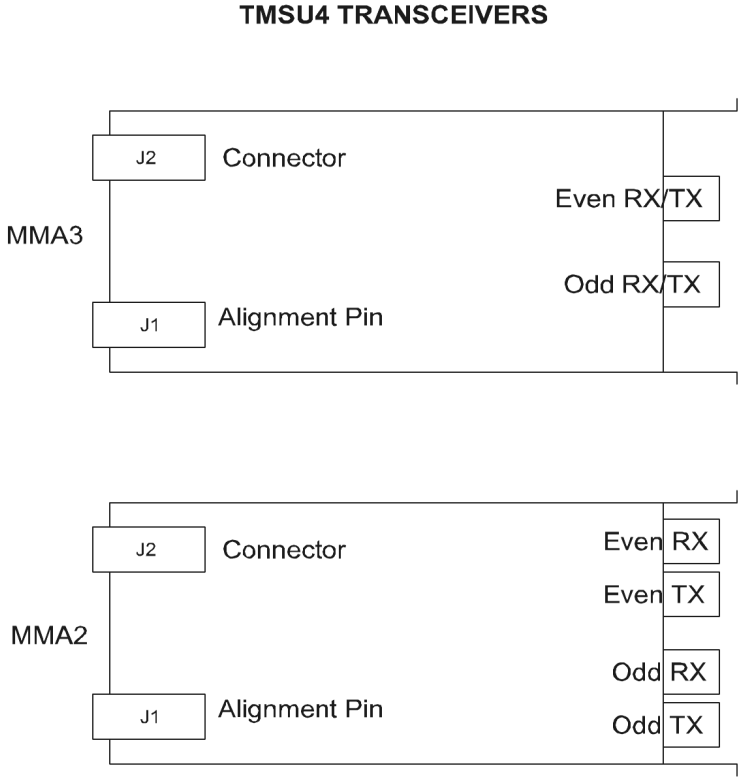


Figure 7. TMSU4 Transceivers

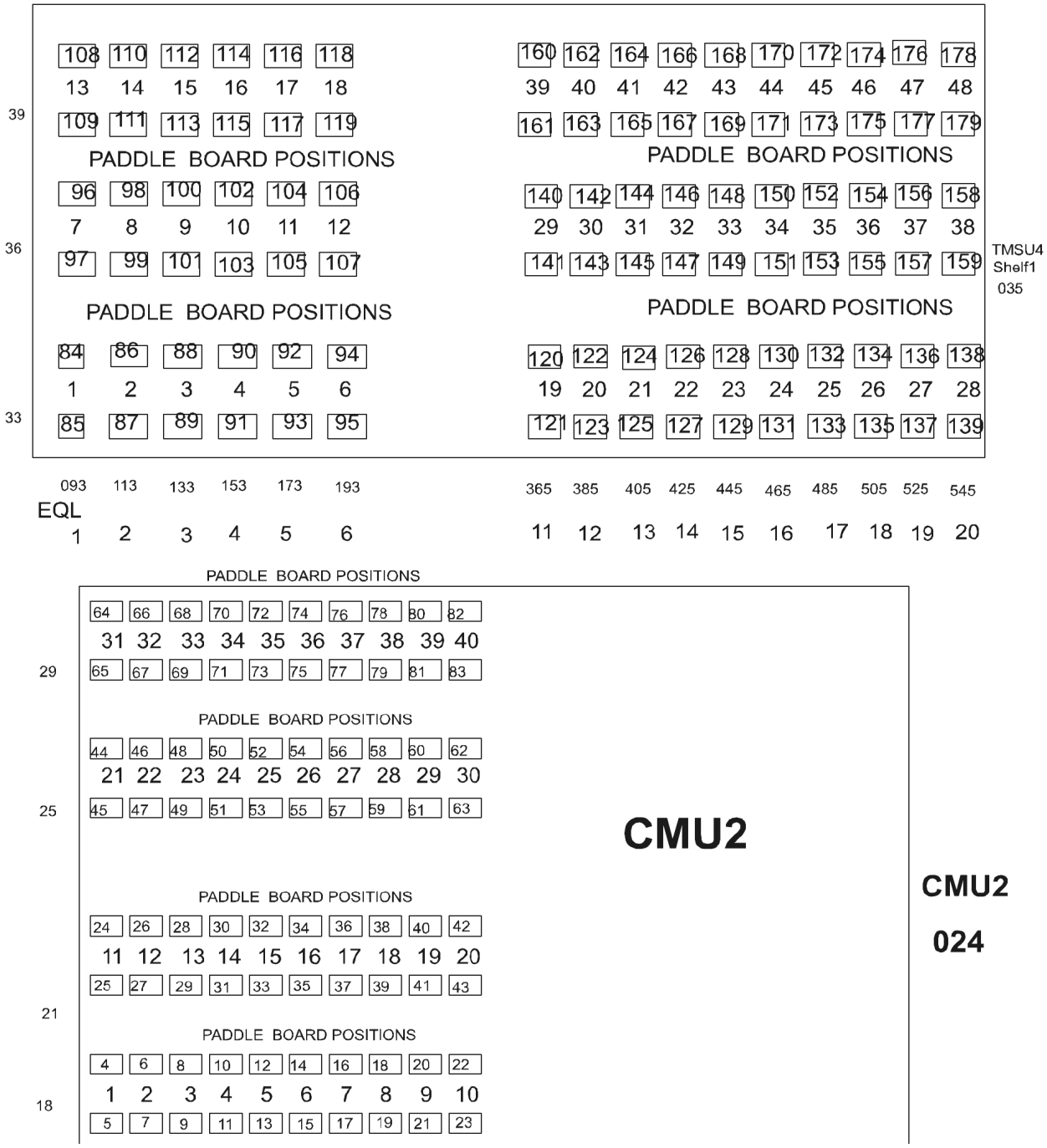


Figure 12. TMSFP 0 CMU2 and TMSU4 Shelf 1

Table

Table C. TMSLNK Migration Order

BLOCK Number =												
NOTE: A "block" in this procedure refers to performing migration of one link set per SM across all affected SMs with the limitation of how many blocks can be migrated in each maintenance window required for the entire job.				**TMSLNK(s) Original**			**Shelf# (Refer to Fig 12-19)**	**TMSLNK(s) New**			**Shelf# (Refer to Fig 12-19)**
				(1201,y,x)				**GRCV 18.16**			
Order: Page Line	**✓**	**SM #**	**LS #**	**(1200,x) NLI Even# NLI Odd#**	**TMSFP #**	**Even Odd**		**TMSFP #**	**Even Odd**		
__-1											
__-2											
__-3											
__-4											
__-5											
__-6											
__-7											
__-8											
__-9											
__-10											
__-11											
__-12											
__-13											
__-14											
__-15											
__-16											